

Supporting Information: Load-Dependent Sacroiliac Joint Micromotion in a Large Pelvic Cohort: Directional Re-routing, Between-Subject Variability and Scaling

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1 Supplementary Methods: implicit mapping, SIJ kinematics, and size scaling

This Supporting Information documents the computational definitions used to (i) represent subject-specific anatomy via an implicit mapping on a common reference domain and to pull back loads defined on the mapped geometry, (ii) extract SIJ kinematics from deformed geometry, and (iii) compute global size proxies and fit size-scaling models.

1.1 Implicit geometry mapping and pullback

For each subject i , the FE solve is performed on a common reference domain Ω_{ref} with coordinates $\mathbf{X}$. Subject-specific geometry is encoded implicitly by a displacement mapping $\phi_i(\mathbf{X})$ such that mapped (subject) coordinates are

$$\mathbf{x} = \chi_i(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X}). \quad (1)$$

In this repository, subject-specific shape information is provided on the template surface as pointwise displacements relative to the template. The mapping field ϕ_i is then constructed by interpolating these displacements to the FE function space (RBF interpolation), yielding an implicit representation of individual geometry on the shared reference mesh. The template (reference) geometry corresponds to $\phi = 0$. Define the deformation gradient and Jacobian of this mapping as

$$\mathbf{F}_i = \mathbf{I} + \nabla \phi_i, \quad J_i = \det(\mathbf{F}_i). \quad (2)$$

The linear elastic displacement field $\mathbf{u}_{i\ell}(\mathbf{X})$ for load case ℓ is solved in reference coordinates using the standard pullback of gradients by $\mathbf{F}_i^{-1}$ (Bonet and Wood, 1997):

$$\boldsymbol{\varepsilon}(\mathbf{w}) = \text{sym}(\nabla \mathbf{w} \mathbf{F}_i^{-1}). \quad (3)$$

Surface measure and traction pullback. Let Γ_{ref} denote a reference boundary region with outward unit normal $\mathbf{n}_{ref}$, and let $\Gamma_i = \chi_i(\Gamma_{ref})$ be the corresponding mapped (subject) surface. The surface measure transforms as

$$dS_x = J_{surf} dS_X, \quad J_{surf} = J_i \|\mathbf{F}_i^{-T} \mathbf{n}_{ref}\|. \quad (4)$$

Therefore, a physical traction $\mathbf{t}_{phys}$ defined on Γ_i contributes

$$\int_{\Gamma_i} \mathbf{t}_{phys} \cdot \mathbf{v} dS_x = \int_{\Gamma_{ref}} \mathbf{t}_{phys} \cdot \mathbf{v} J_{surf} dS_X. \quad (5)$$

Total-force standardization. To impose a prescribed resultant force vector $\mathbf{F}^{tot}$ on the mapped surface, a constant traction is chosen as

$$\mathbf{t}_{phys} = \frac{\mathbf{F}^{tot}}{A_{phys}}, \quad A_{phys} = \int_{\Gamma_{ref}} J_{surf} dS_X, \quad (6)$$

which ensures $\int_{\Gamma_i} \mathbf{t}_{phys} dS_x = \mathbf{F}^{tot}$. (Pressure loads use the corresponding Piola transform $J_i \mathbf{F}_i^{-T} \mathbf{n}_{ref}$.)

Deformed coordinates for point clouds. For a reference point $\mathbf{X}$ on the template surface, the deformed coordinate used for kinematics is

$$\mathbf{x}_{i\ell}(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X}) + \mathbf{u}_{i\ell}(\mathbf{X}). \quad (7)$$

In practice, ϕ_i and $\mathbf{u}_{i\ell}$ are evaluated at surface points by interpolation from the FE degrees of freedom. In the present workflow, this evaluation uses radial basis function (RBF) interpolation constructed on FE degree-of-freedom coordinates.

1.2 Notation and inputs

For each subject i and load case ℓ , the finite-element solution defines a deformed pelvis geometry obtained by evaluating $\mathbf{x}_{i\ell}(\mathbf{X})$ on the template surface points (one-to-one correspondence). In this repository, the reference surface mesh (“ref”) is the common template in its reference configuration, and the deformed surface mesh (“def”) is obtained by applying the subject mapping ϕ_i and the load-specific displacement $\mathbf{u}_{i\ell}$ to the same point set. After splitting the surface into bodies (left ilium, right ilium, sacrum), denote the reference point clouds as $\{\mathbf{X}_j^{LI}\}$, $\{\mathbf{X}_j^{RI}\}$, and $\{\mathbf{X}_j^S\}$, and the corresponding deformed point clouds as $\{\mathbf{x}_j^{LI}\}$, $\{\mathbf{x}_j^{RI}\}$, and $\{\mathbf{x}_j^S\}$.

All vectors are 3D column vectors. Rotation matrices are 3×3 and are orthogonal with determinant $+1$. Translations are 3D vectors. Angles are reported in degrees and translations in mm.

1.3 Pelvis-aligned coordinate frame (ML/AP/CC)

All SIJ outputs are expressed in a pelvis-aligned orthonormal frame whose columns are the unit axes $\hat{\mathbf{x}}$ (ML), $\hat{\mathbf{y}}$ (AP), and $\hat{\mathbf{z}}$ (CC). The frame is computed from the reference (undeformed) pelvis point cloud.

Origin. Let $\mathbf{o}$ be the centroid of all reference surface points:

$$\mathbf{o} = \frac{1}{N} \sum_{j=1}^N \mathbf{X}_j. \quad (8)$$

ML axis. Compute the sample covariance matrix of the centered point cloud and its eigen-decomposition. Let $\hat{\mathbf{x}}$ be the eigenvector associated with the largest eigenvalue (the dominant PCA axis), oriented so that its global x component is nonnegative (a deterministic sign convention):

$$\hat{\mathbf{x}} = \arg \max_{\|\mathbf{v}\|=1} \mathbf{v}^T \text{cov}(\mathbf{X}_j - \mathbf{o}) \mathbf{v}. \quad (9)$$

CC and AP axes. Project the centered point cloud into the plane orthogonal to $\hat{\mathbf{x}}$:

$$\tilde{\mathbf{X}}_j = (\mathbf{X}_j - \mathbf{o}) - ((\mathbf{X}_j - \mathbf{o}) \cdot \hat{\mathbf{x}})\hat{\mathbf{x}}. \quad (10)$$

The dominant direction in this plane is extracted by PCA/SVD and then orthogonalized against $\hat{\mathbf{x}}$ to obtain $\hat{\mathbf{z}}$. The remaining axis is defined to ensure a right-handed frame:

$$\hat{\mathbf{y}} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}, \quad \hat{\mathbf{z}} = \frac{\hat{\mathbf{x}} \times \hat{\mathbf{y}}}{\|\hat{\mathbf{x}} \times \hat{\mathbf{y}}\|}. \quad (11)$$

Left-right sign convention. To enforce that positive ML points from left to right, the ML axis is flipped (with the AP axis simultaneously flipped to preserve right-handedness) if the left ilium mean projects more positively than the right ilium mean along $\hat{\mathbf{x}}$:

$$(\bar{\mathbf{X}}^{LI} - \mathbf{o}) \cdot \hat{\mathbf{x}} > (\bar{\mathbf{X}}^{RI} - \mathbf{o}) \cdot \hat{\mathbf{x}} \Rightarrow \hat{\mathbf{x}} \leftarrow -\hat{\mathbf{x}}, \hat{\mathbf{y}} \leftarrow -\hat{\mathbf{y}}. \quad (12)$$

Collect axes into the orthonormal frame matrix

$$\mathbf{P} = \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \end{bmatrix}, \quad (13)$$

which maps world vectors into pelvis coordinates by $\mathbf{v}_{pel} = \mathbf{P}^\top \mathbf{v}$.

1.4 Removing global affine motion using the sacrum

To focus on SIJ-relative motion, any global affine deformation of the entire pelvis is removed using a sacrum-based fit. Let $\mathbf{X}_j^S$ and $\mathbf{x}_j^S$ be corresponding sacrum points in reference and deformed meshes. We estimate an affine map $(\mathbf{A}, \mathbf{b})$ by least squares:

$$(\mathbf{A}, \mathbf{b}) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_j \|\mathbf{A}\mathbf{X}_j^S + \mathbf{b} - \mathbf{x}_j^S\|^2. \quad (14)$$

The inverse affine map is then applied to all deformed points (sacrum and both ilia) to obtain affine-corrected deformed coordinates:

$$\mathbf{x}' = \mathbf{A}^+ (\mathbf{x} - \mathbf{b}), \quad (15)$$

where $\mathbf{A}^+$ denotes the Moore–Penrose pseudoinverse (robust to mild ill-conditioning).

1.5 Defining a side-specific SIJ region of interest (ROI)

SIJ rigid fits are computed over surface ROIs intended to approximate the articular contact regions. On each side $s \in \{L, R\}$ we first select candidate points by the sign of the ML coordinate in the reference frame:

$$s = L : (\mathbf{X} - \mathbf{o}) \cdot \hat{\mathbf{x}} < 0, \quad s = R : (\mathbf{X} - \mathbf{o}) \cdot \hat{\mathbf{x}} > 0. \quad (16)$$

Within each side, ROIs are defined by proximity between the sacrum and ilium surfaces in the reference configuration. For a sacrum point $\mathbf{p}$, define its distance to the opposite bone surface as

$$d(\mathbf{p}, \mathcal{B}) = \min_{\mathbf{q} \in \mathcal{B}} \|\mathbf{p} - \mathbf{q}\|. \quad (17)$$

Given a gap tolerance $g = 5$ mm, the sacral ROI on side s is

$$\mathcal{S}_s = \{\mathbf{X}_j^S : d(\mathbf{X}_j^S, \{\mathbf{X}_k^{I_s}\}) \leq g\}, \quad (18)$$

and the corresponding iliac ROI is

$$\mathcal{I}_s = \{\mathbf{X}_k^{I_s} : d(\mathbf{X}_k^{I_s}, \{\mathbf{X}_j^S\}) \leq g\}. \quad (19)$$

If either ROI contains fewer than $N_{min} = 80$ points, the N_{min} closest points by $d(\cdot)$ are used instead (a deterministic fallback to avoid empty/small ROIs).

1.6 Rigid-body fit on each bone ROI (trimmed Kabsch)

For each side s and each bone (sacrum and ilium), a rigid transform $(\mathbf{R}, \mathbf{t})$ is fit that maps reference ROI points to affine-corrected deformed ROI points. Given corresponding point sets $\{\mathbf{p}_k\}$ and $\{\mathbf{q}_k\}$, we solve

$$(\mathbf{R}, \mathbf{t}) = \arg \min_{\mathbf{R} \in SO(3), \mathbf{t}} \sum_k \|\mathbf{R}\mathbf{p}_k + \mathbf{t} - \mathbf{q}_k\|^2. \quad (20)$$

The Kabsch solution is used. Let $\bar{\mathbf{p}}$ and $\bar{\mathbf{q}}$ be the point-set centroids and define centered coordinates $\mathbf{p}_k^c = \mathbf{p}_k - \bar{\mathbf{p}}$, $\mathbf{q}_k^c = \mathbf{q}_k - \bar{\mathbf{q}}$. The cross-covariance is

$$\mathbf{C} = \sum_k \mathbf{p}_k^c (\mathbf{q}_k^c)^\top. \quad (21)$$

With SVD $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, the rotation is

$$\mathbf{R} = \mathbf{V}\mathbf{D}\mathbf{U}^\top, \quad \mathbf{D} = \text{diag}(1, 1, \det(\mathbf{V}\mathbf{U}^\top)), \quad (22)$$

and the translation is

$$\mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}. \quad (23)$$

To reduce sensitivity to outliers, the fit is *trimmed*: residuals $e_k = \|\mathbf{R}\mathbf{p}_k + \mathbf{t} - \mathbf{q}_k\|$ are computed, the smallest $(1 - \tau)$ fraction is retained ($\tau = 0.1$), and the rigid fit is refit on the retained subset. This trimming-refit procedure is run for two iterations.

1.7 Relative SIJ rotation and translation

Let $(\mathbf{R}_s^S, \mathbf{t}_s^S)$ and $(\mathbf{R}_s^I, \mathbf{t}_s^I)$ be the rigid transforms for the sacrum and the ilium on side s . The ilium motion relative to the sacrum is expressed by

$$\mathbf{R}_s^{rel} = (\mathbf{R}_s^S)^\top \mathbf{R}_s^I, \quad \mathbf{t}_s^{rel} = (\mathbf{R}_s^S)^\top (\mathbf{t}_s^I - \mathbf{t}_s^S). \quad (24)$$

Euler angles (NUT/AP/CC). To report rotations about the pelvis axes, we express the relative rotation in the pelvis frame:

$$\mathbf{R}_s^{pel} = \mathbf{P}^\top \mathbf{R}_s^{rel} \mathbf{P}. \quad (25)$$

Euler XYZ angles are then extracted as

$$\alpha_x = \text{atan2}(\mathbf{R}_{32}^{pel}, \mathbf{R}_{33}^{pel}), \quad \beta_y = \arcsin(-\mathbf{R}_{31}^{pel}), \quad \gamma_z = \text{atan2}(\mathbf{R}_{21}^{pel}, \mathbf{R}_{11}^{pel}), \quad (26)$$

with standard singularity handling when $|\mathbf{R}_{31}^{pel}| \approx 1$ (gimbal-lock case). Angles are converted to degrees and reported as NUT (α_x), AP-axis rotation (β_y), and CC-axis rotation (γ_z).

Translation (ML/AP/CC). We define a representative SIJ contact point as the centroid of the sacral ROI in the *reference* configuration, $\mathbf{p}_s = \text{mean}(\mathcal{S}_s)$. The relative displacement between ilium and sacrum at this point (in world coordinates) is computed as

$$\mathbf{d}_s^{world} = (\mathbf{R}_s^I - \mathbf{R}_s^S)\mathbf{p}_s + (\mathbf{t}_s^I - \mathbf{t}_s^S), \quad (27)$$

and expressed in pelvis coordinates by

$$\mathbf{d}_s^{pel} = \mathbf{P}^\top \mathbf{d}_s^{world} = \begin{bmatrix} d_{ML} \\ d_{AP} \\ d_{CC} \end{bmatrix}_s. \quad (28)$$

These components are reported (mm) as dx_{ML} , dy_{AP} , and dz_{CC} for the left and right SIJ.

1.8 Derived subject-level SIJ metrics used in the manuscript

For each subject i , load ℓ , and side $s \in \{L, R\}$, define the rotation and translation component vectors

$$\boldsymbol{\theta}_{i\ell s} = \begin{bmatrix} \alpha_x \\ \beta_y \\ \gamma_z \end{bmatrix}_{i\ell s}, \quad \mathbf{d}_{i\ell s} = \begin{bmatrix} d_{ML} \\ d_{AP} \\ d_{CC} \end{bmatrix}_{i\ell s}. \quad (29)$$

The side-specific magnitudes are the Euclidean norms

$$r_{i\ell s} = \|\boldsymbol{\theta}_{i\ell s}\|_2, \quad t_{i\ell s} = \|\mathbf{d}_{i\ell s}\|_2. \quad (30)$$

Primary outcomes are the bilateral means of these magnitudes:

$$\text{rot_mag_deg}_{i\ell} = \frac{1}{2}(r_{i\ell L} + r_{i\ell R}), \quad \text{trans_mag_mm}_{i\ell} = \frac{1}{2}(t_{i\ell L} + t_{i\ell R}). \quad (31)$$

Directional component summaries are bilateral means of absolute component magnitudes, e.g.

$$\text{ap_trans_abs_mm}_{i\ell} = \frac{1}{2}(|d_{AP,i\ell L}| + |d_{AP,i\ell R}|), \quad (32)$$

with analogous definitions for ML and CC translation and for the rotational components (nutation/AP/CC). Left–right asymmetry magnitudes are defined on the scalar magnitudes:

$$\text{lr_rot_asym_deg}_{i\ell} = |r_{i\ell L} - r_{i\ell R}|, \quad \text{lr_trans_asym_mm}_{i\ell} = |t_{i\ell L} - t_{i\ell R}|. \quad (33)$$

1.9 Global size proxies and size-scaling model

Mapped volume, surface area, and scale. Global size proxies were computed from the mapped pelvis geometry implied by ϕ_i . Using the implicit mapping χ_i , the mapped total pelvic volume V_i (cm³) and total pelvic surface area S_i (cm²) can be expressed as

$$V_i = \int_{\Omega_{ref}} J_i(\mathbf{X}) d\mathbf{X}, \quad S_i = \int_{\Gamma_{ref}} J_{surf}(\mathbf{X}) dS_X, \quad (34)$$

with $J_i = \det(\mathbf{F}_i)$ and J_{surf} defined above. An isotropic global scale factor is defined by

$$\text{scale}_i = \left(\frac{V_i}{V_{template}} \right)^{1/3}, \quad (35)$$

where $V_{template}$ is the total volume of the template pelvis.

Size-scaling model. For each load case ℓ and each primary SIJ outcome $Y_{i\ell}$ (rotation magnitude and translation magnitude), we fit a log–log model

$$\begin{aligned} \log(Y_{i\ell}) = & \beta_0 + \beta_1 \log(V_i) + \beta_2 \text{sex}_i + \beta_3 \text{age}_{z,i} + \beta_4 \text{AP}_{z,i} + \beta_5 \text{Biischadic}_{z,i} + \beta_6 \text{Subpubic}_{z,i} \\ & + \beta_7 \log(V_i) \text{sex}_i + \varepsilon_{i\ell}, \end{aligned} \quad (36)$$

where $\text{sex}_i = 1$ for female and 0 for male, and x_z denotes z -scoring $x_z = (x - \bar{x})/s_x$ using the sample standard deviation ($ddof = 1$). Models are estimated by OLS with HC3 heteroskedasticity-consistent covariance.

Exponent extraction and confidence intervals. In the above parameterization, the male scaling exponent is $b_M = \beta_1$ and the female scaling exponent is $b_F = \beta_1 + \beta_7$. Wald-type 95% CIs use the model covariance matrix:

$$b_M \pm 1.96\sqrt{\text{Var}(\beta_1)}, \quad b_F \pm 1.96\sqrt{\text{Var}(\beta_1) + \text{Var}(\beta_7) + 2\text{Cov}(\beta_1, \beta_7)}. \quad (37)$$

The sex-by-size slope interaction is tested by the p value of β_7 , with Benjamini–Hochberg false-discovery control applied across the interaction tests.

Table S1: Directional re-routing metrics

load_case	metric	median_delta	ci95_low	ci95_high	p_value	cohen_dz	p_fdr_bh
SP1leg	ap_trans_abs_mm	0.072	0.045	0.096	0.000	0.681	0.000
SP1leg	ml_trans_abs_mm	0.005	-0.004	0.015	0.095	0.182	0.114
SP1leg	cc_trans_abs_mm	0.394	0.347	0.434	0.000	1.211	0.000
LAB_phase1	ap_trans_abs_mm	0.015	-0.037	0.050	0.840	-0.020	0.840
LAB_phase1	ml_trans_abs_mm	0.086	0.077	0.095	0.000	0.978	0.000
LAB_phase1	cc_trans_abs_mm	-0.850	-0.881	-0.811	0.000	-2.864	0.000
LAB_phase2	ap_trans_abs_mm	0.163	0.102	0.213	0.000	0.345	0.000
LAB_phase2	ml_trans_abs_mm	0.145	0.132	0.160	0.000	1.200	0.000
LAB_phase2	cc_trans_abs_mm	-0.903	-0.931	-0.853	0.000	-2.617	0.000
LAB_phase3	ap_trans_abs_mm	-0.001	-0.012	0.015	0.493	0.059	0.538
LAB_phase3	ml_trans_abs_mm	-0.041	-0.052	-0.033	0.000	-0.788	0.000
LAB_phase3	cc_trans_abs_mm	-0.586	-0.609	-0.544	0.000	-3.116	0.000

Table S2: Variance-channel decomposition

load_case	metric	var_full	var_shape_only	var_material_only	shape_over_full_pct	shape_over_full_ci95_low	shape_over_full_ci95_high	material_over_full_pct	material_over_full_ci95_low	material_over_full_ci95_high
SP2leg	rot_mag_deg	0.764	0.761	0.000	99.577	99.018	100.165	0.053	0.039	0.072
SP2leg	trans_mag_mm	0.173	0.173	0.001	100.072	98.729	101.358	0.295	0.221	0.389
SP2leg	ap_trans_abs_mm	0.053	0.054	0.000	102.140	100.142	105.016	0.122	0.079	0.187
SP2leg	ml_trans_abs_mm	0.070	0.070	0.000	100.462	99.780	101.181	0.067	0.050	0.089
SP2leg	cc_trans_abs_mm	0.158	0.158	0.000	99.857	98.427	101.276	0.315	0.239	0.411
SP1leg	rot_mag_deg	0.643	0.646	0.000	100.388	99.718	101.039	0.033	0.023	0.048
SP1leg	trans_mag_mm	0.241	0.243	0.000	100.860	99.700	102.073	0.086	0.061	0.121
SP1leg	ap_trans_abs_mm	0.049	0.050	0.000	100.731	98.906	102.774	0.317	0.228	0.440
SP1leg	ml_trans_abs_mm	0.060	0.061	0.000	100.694	100.002	101.426	0.005	0.003	0.008
SP1leg	cc_trans_abs_mm	0.201	0.203	0.000	100.799	99.499	102.268	0.164	0.120	0.224
LAB_phase1	rot_mag_deg	0.630	0.625	0.000	99.246	98.715	99.799	0.030	0.023	0.039
LAB_phase1	trans_mag_mm	0.099	0.100	0.000	100.758	99.909	101.684	0.049	0.038	0.062
LAB_phase1	ap_trans_abs_mm	0.043	0.043	0.000	100.558	99.632	101.608	0.048	0.036	0.064
LAB_phase1	ml_trans_abs_mm	0.086	0.086	0.000	100.096	99.835	100.377	0.008	0.006	0.011
LAB_phase1	cc_trans_abs_mm	0.042	0.042	0.000	100.600	98.948	102.541	0.272	0.207	0.356
LAB_phase2	rot_mag_deg	0.604	0.596	0.000	98.696	97.875	99.496	0.003	0.002	0.005
LAB_phase2	trans_mag_mm	0.120	0.121	0.000	100.780	100.065	101.497	0.008	0.007	0.011
LAB_phase2	ap_trans_abs_mm	0.079	0.080	0.000	100.702	99.924	101.387	0.014	0.011	0.019
LAB_phase2	ml_trans_abs_mm	0.091	0.092	0.000	100.169	99.810	100.564	0.013	0.009	0.019
LAB_phase2	cc_trans_abs_mm	0.029	0.030	0.000	101.928	99.736	104.281	0.067	0.041	0.104
LAB_phase3	rot_mag_deg	0.955	0.963	0.000	100.795	100.324	101.276	0.002	0.001	0.003
LAB_phase3	trans_mag_mm	0.101	0.100	0.000	99.713	99.353	100.130	0.016	0.012	0.022
LAB_phase3	ap_trans_abs_mm	0.061	0.061	0.000	99.708	99.114	100.426	0.033	0.023	0.050
LAB_phase3	ml_trans_abs_mm	0.058	0.058	0.000	99.770	99.574	99.991	0.001	0.001	0.002
LAB_phase3	cc_trans_abs_mm	0.086	0.086	0.000	99.529	98.971	100.025	0.009	0.007	0.012

Table S3: LAB-phase sex-adjusted models

load_case	outcome	beta_female_minus_male	ci95_low	ci95_high	p_value	beta_std	partial_r2	p_fdr_bh
LAB_phase1	rot_mag_deg	0.396	0.009	0.783	0.045	0.428	0.015	0.054
LAB_phase1	trans_mag_mm	0.229	0.078	0.379	0.003	0.405	0.032	0.007
LAB_phase2	rot_mag_deg	0.281	-0.126	0.688	0.177	0.303	0.007	0.177
LAB_phase2	trans_mag_mm	0.199	0.039	0.359	0.015	0.353	0.022	0.022
LAB_phase3	rot_mag_deg	0.627	0.206	1.048	0.004	0.677	0.031	0.007
LAB_phase3	trans_mag_mm	0.233	0.078	0.388	0.003	0.413	0.031	0.007

Table S4: Size-scaling sensitivity using log(scale)

load_case	outcome	male_exponent_scale	male_ci95_low	male_ci95_high	female_exponent_scale	female_ci95_low	female_ci95_high	sex_slope_interaction_beta_scale	sex_slope_interaction_p_scale	adj_r2	n	interaction_p_fdr_bh_scale
SP2leg	rot_mag_deg	-1.001	-2.793	0.790	-0.917	-2.285	0.452	0.085	0.934	0.383	278	0.958
SP2leg	trans_mag_mm	-0.682	-1.685	0.321	-0.632	-1.515	0.252	0.050	0.935	0.468	278	0.958
SP1leg	rot_mag_deg	-1.447	-2.611	-0.283	-1.283	-2.286	0.164	0.164	0.812	0.382	278	0.958
SP1leg	trans_mag_mm	-1.708	-2.607	-0.808	-1.296	-2.008	-0.583	0.412	0.436	0.500	278	0.958
LAB_phase1	rot_mag_deg	-0.515	-2.537	1.507	-1.248	-2.878	0.382	-0.733	0.544	0.267	278	0.958
LAB_phase1	trans_mag_mm	-0.530	-1.994	0.934	-1.256	-2.743	0.230	-0.726	0.414	0.354	278	0.958
LAB_phase2	rot_mag_deg	-0.683	-3.061	1.694	-1.002	-2.674	0.670	-0.319	0.803	0.269	278	0.958
LAB_phase2	trans_mag_mm	-1.100	-2.712	0.512	-1.841	-3.184	-0.499	-0.741	0.430	0.375	278	0.958
LAB_phase3	rot_mag_deg	-1.469	-3.508	0.523	-1.555	-3.143	0.033	-0.862	0.958	0.367	278	0.958
LAB_phase3	trans_mag_mm	-0.659	-2.159	0.840	-0.812	-2.123	0.499	-0.152	0.864	0.310	278	0.958

References

Bonet, J. and Wood, R. D. (1997). *Nonlinear Continuum Mechanics for Finite Element Analysis*. Cambridge University Press, 1st edition.